

BIG DATA SYSTEMS • CC ZG522

Hadoop-Based Sales Analytics for Business Performance Analysis

Situated Learning – Project Proposal

Apurbaa Deb

2024dc04149@wilp.bits-pilani.ac.in

11th August 2026

HDFS

YARN

MapReduce

Sales Data Is Growing Faster Than Manual Analysis Can Handle

Sales transactions carry rich signal — customers, products, regions, channels, payments — but as volume grows, conventional tools can't scale to analyze them efficiently. A scalable Big Data platform is needed to turn 10,000+ raw transactions into trend, regional, and channel-level insight.



Volume

10,000 transactions in the PoC; production systems generate millions over time.



Velocity

Continuous transactions across Online, Store & Distributor channels demand regular processing.



Variety

Dates, IDs, categories, regions, payments & channels — mixed structured data types.

Core Objective

How can a scalable Big Data platform analyze sales transactions and identify monthly trends, and performance across regions, categories, and channels — to support business decision-making?

Three Hadoop Components Power the Solution

HDFS stores the data, YARN manages resources, and MapReduce processes it — together forming the analytics engine.



HDFS

Hadoop Distributed File System

Stores large datasets across the cluster. sales.csv is uploaded to HDFS so Hadoop can process it, and job outputs are written back to HDFS.

```
hdfs dfs -put sales.csv /sales/input/
```



YARN

Yet Another Resource Negotiator

Manages cluster resources and job scheduling. When a MapReduce job runs, YARN allocates CPU & memory, starts processes, and monitors execution.

```
Allocates → Starts → Monitors →  
Coordinates
```



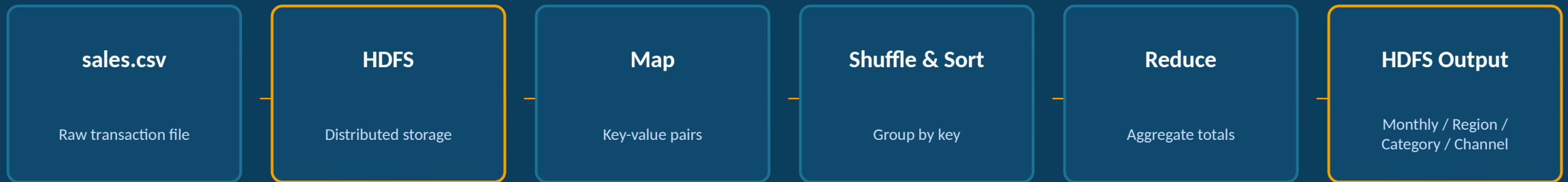
MapReduce

Distributed Processing Model

Processes records in Map → Shuffle/Sort → Reduce stages, turning raw rows into aggregated key-value results.

```
Jan→[12500,8300] → Jan: 20800
```

End-to-End Processing Workflow



YARN manages resources & scheduling for every job across this pipeline

Worked Example — Monthly Sales

Map

January → 12500
January → 8300
February → 15200

Shuffle & Sort

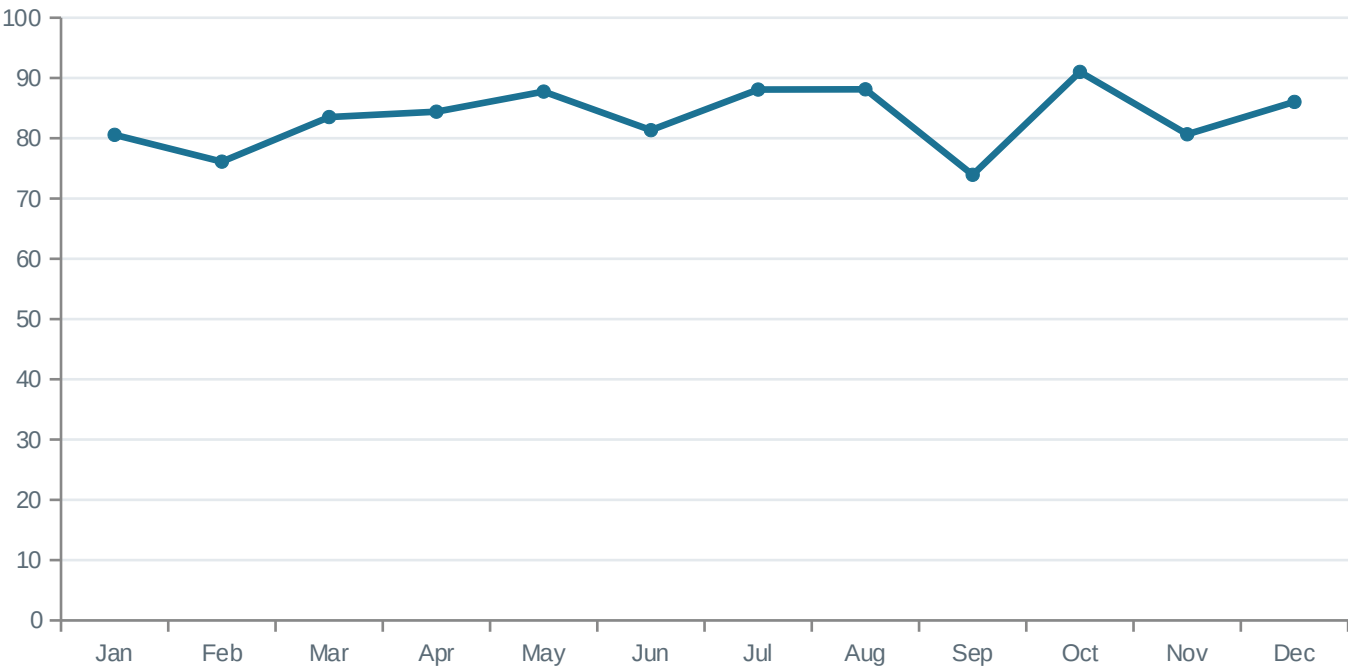
→ January → [12500, 8300]
February → [15200]

Reduce

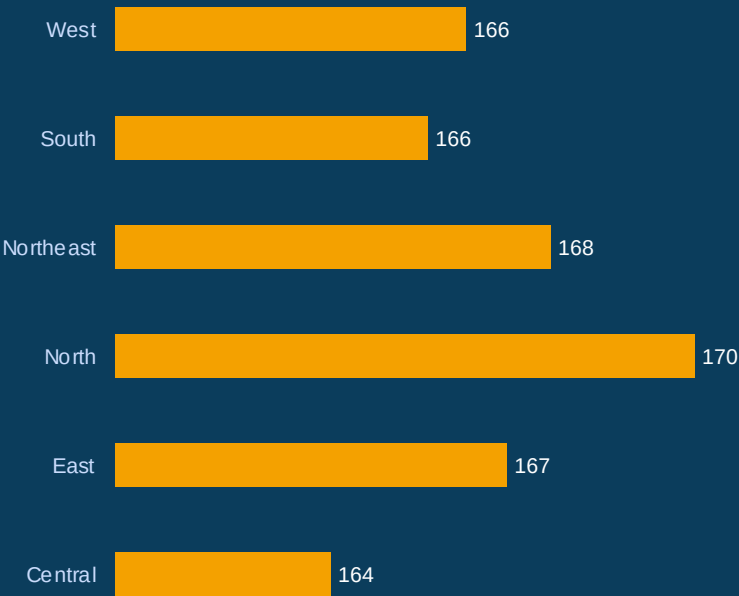
→ January → 20800
February → 15200

Monthly & Regional Sales — 10,000 Transactions Processed

Monthly Sales Trend (₹ Millions)



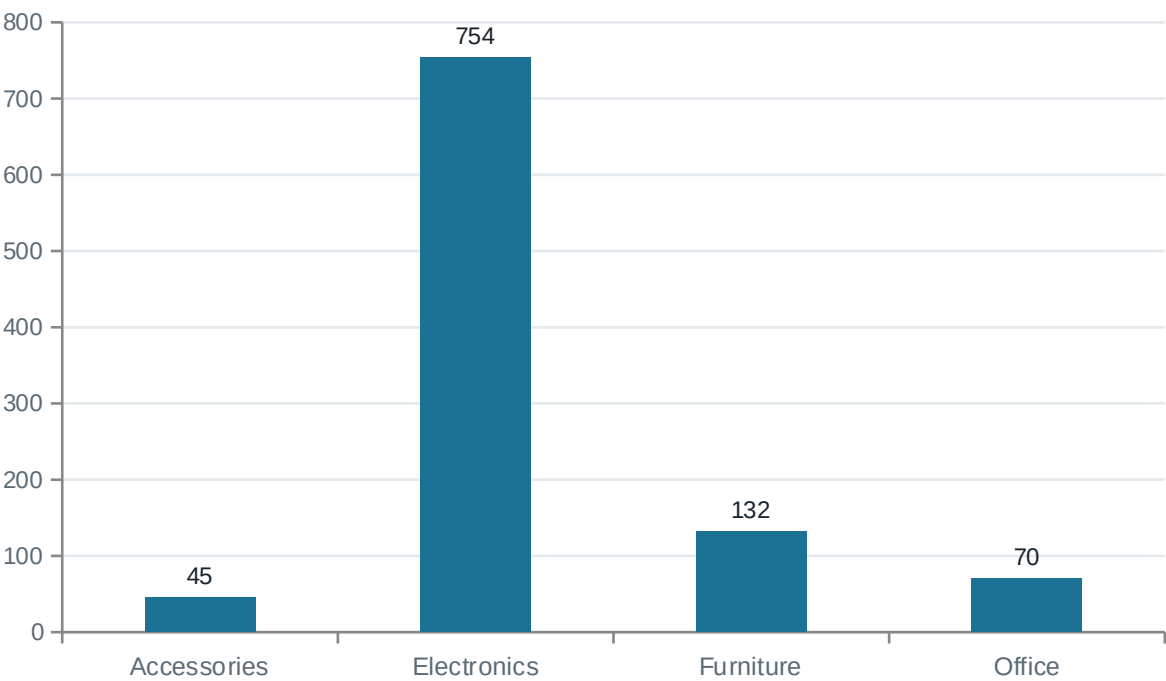
Regional Sales (₹M)



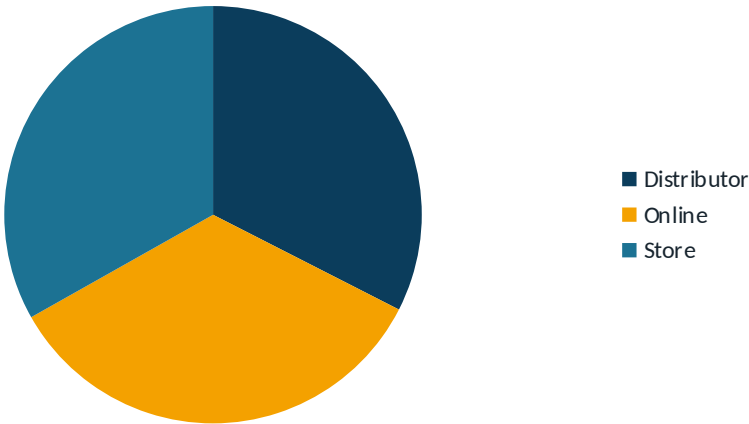
Highest: October 2026 (₹91.01M) • Lowest: September 2026 (₹73.94M) • Top region: North (₹170.01M)

Electronics & Online Lead Performance

Category Sales (₹ Millions)



Channel Sales (₹M)



753.87M
343.68M

Electronics — dominant category
Online — top-performing channel

From Proof of Concept to Production Impact

Business Impact

- Reduces manual effort to process & aggregate large sales datasets
- Monthly trends support sales planning & forecasting
- Regional analysis reveals geographic performance gaps
- Category insight guides inventory & demand planning
- Channel comparison informs Online vs. Store vs. Distributor strategy

Future Scope

- Automated data ingestion & scheduled processing
- Scale to millions of transactions
- Data-quality validation checks
- Additional KPIs — average order value, growth rate
- Business dashboards for visualization
- Near-real-time & predictive sales analytics